

Synthetic Function and the Clotting System.

When a child with liver disease has blood taken, one of the first results to come back is the INR. It is often the number that frightens people most. A resident sees an INR of two and asks for plasma before a line is placed. A surgeon delays a biopsy until the number comes down. A team in intensive care watches it hour by hour. Yet the INR is one of the most misunderstood tests in hepatology. It is an excellent measure of how sick the liver is, and a poor measure of whether the child will bleed.

This chapter is about what the liver makes, and about how to read the tests that measure it. We will start with the map: which proteins come from the liver cell, which come from elsewhere, and how quickly each one disappears. Then we will look at the INR itself, why it was built, and why the same blood can give two different results. After that we will learn how to separate the three common causes of a long INR: vitamin K deficiency, liver failure and consumption. We will spend some time on vitamin K in infants, because this is where paediatric hepatology differs most from adult practice. Finally we will look at the modern idea of rebalanced haemostasis, at what plasma really does, at who actually bleeds during procedures, and at what the major guidelines say.

A note on units. Clotting factor levels are given as a percentage of normal adult plasma. Fibrinogen is given in milligrams per decilitre. Platelet counts are given in thousands per microlitre, so a platelet count of fifty means fifty thousand per microlitre, which is the same as fifty times ten to the ninth per litre.

Part one. What the liver makes.

The clotting system has four parts, and the liver makes a large share of every one of them. This is the most important idea in the chapter, so it is worth going slowly.

The first part is the clotting cascade itself, the procoagulant factors. The liver cell makes fibrinogen and factors two, five, seven, nine, ten, eleven and thirteen. Factor two is also called prothrombin. There is one important exception. Factor eight is made mainly outside the liver cell, by the endothelial cells that line the blood vessels, including the sinusoids of the liver. So when the liver fails, the hepatic factors fall, but factor eight does not. In fact, it usually rises.

The second part is the natural anticoagulants, the brakes on the system. The liver makes antithrombin, protein C and protein S. These fall in liver disease in step with the procoagulant factors. One partner of this system, thrombomodulin, sits

on the surface of the endothelium. Thrombomodulin is what switches protein C on. Remember this, because it explains why the INR misleads us.

The third part is primary haemostasis, the platelet plug. The liver makes thrombopoietin, the hormone that drives platelet production. The stellate cells of the liver make ADAMTS thirteen, the enzyme that cuts von Willebrand factor into smaller, less sticky pieces. Von Willebrand factor itself comes from endothelium. So in liver disease the platelet count falls, because thrombopoietin is low and the spleen is large, while von Willebrand factor rises and ADAMTS thirteen falls.

The fourth part is fibrinolysis, the system that dissolves clots. The liver makes plasminogen, alpha two antiplasmin and a protein called TAFI, which stands for thrombin-activatable fibrinolysis inhibitor. Endothelium makes tissue plasminogen activator and its inhibitor, PAI one. In liver disease, proteins on both sides fall, and the net effect differs from one patient to another.

There is one more step that belongs to the liver. Factors two, seven, nine and ten, and proteins C and S, are all made in an unfinished form. An enzyme in the liver cell, the vitamin K-dependent carboxylase, adds a final chemical group to each of them. Without it, they are released into the blood but cannot work. This enzyme needs vitamin K. Vitamin K comes from the diet and from gut bacteria, and because it is fat soluble, it needs bile in the gut to be absorbed. So a child with cholestasis can run out of vitamin K and develop a long INR while every liver cell is still working perfectly well. The liver has not failed. It has simply been starved.

Keep this map in mind. Almost everything else in this chapter follows from it. The liver makes the accelerator and the brakes. It makes the clotting factors and the clot-dissolving proteins. And some of its products depend on bile flow as much as on the health of the liver cell.

Part two. What falls first.

Clotting factors do not all disappear from the blood at the same speed. Each has its own half-life, and that decides the order in which they fall when the liver stops making them.

Factor seven has the shortest half-life, about five hours. Factor eight is about ten hours. Factors nine and five are around a day. Factor ten is about forty hours. Prothrombin is about two and a half days. Fibrinogen lasts three to five days.

The prothrombin time, and therefore the INR, is driven mainly by factor seven. Because factor seven disappears within hours, the INR moves within hours of any change in the liver's ability to make proteins. That makes it a very fast signal. Albumin, by contrast, has a half-life of about three weeks. A child whose liver

fails suddenly may have a normal albumin for days while the INR is already climbing.

So think of the INR as the liver's fast clock, and albumin as its slow clock. The INR tells you what the liver is doing today. The albumin tells you what it has been doing over the past weeks. Read both before you decide whether a child is getting better or worse.

Part three. The newborn is different.

Before we can judge an INR in a baby, we need to know what is normal for a baby. The classic study was done by Maureen Andrew and colleagues in Hamilton, Canada, and published in 1987. They measured the clotting system in one hundred and eighteen healthy term infants.

On the first day of life, the vitamin K-dependent factors were about half of adult values. Prothrombin was about forty-eight percent of the adult level, and factor seven about sixty-six percent. Factor five was about seventy-two percent. But the natural anticoagulants were low as well. Protein C was about thirty-five percent of the adult level, protein S about thirty-six percent, and antithrombin about sixty-three percent. Factor eight was already at adult levels, and von Willebrand factor was about one and a half times the adult level.

So a healthy newborn is not a small adult with a clotting problem. The newborn is balanced, but at a lower level, with low accelerators and low brakes, and a strong push from von Willebrand factor. Healthy newborns do not bleed more easily than adults. The prothrombin time on day one was thirteen seconds, compared with about twelve and a half seconds at six months of age.

The practical lesson is simple. Judge a newborn's INR against newborn reference ranges, ideally from your own laboratory and your own analyser. Never judge it against adult cut-offs.

Part four. A test built for warfarin.

The prothrombin time has an interesting history. It was first developed to study bleeding in patients with obstructive jaundice, which we now know was vitamin K deficiency. Later it became the test used to monitor warfarin. The INR, or international normalised ratio, was created so that a warfarin clinic in one city could compare its results with a clinic in another.

To make that work, each batch of laboratory reagent is given a number called the ISI, the international sensitivity index. The ISI is calibrated using plasma from patients who are taking warfarin or a similar drug. So the INR is standardised for warfarin, not for liver disease. In a patient with cirrhosis or liver failure, the same

blood can give quite different INR values depending on which reagent the laboratory uses.

The test also has a basic design limit. It measures how long it takes plasma to form a clot after tissue factor is added. The clock stops when less than five percent of the thrombin has been made. So it looks only at the very first moments of clotting. It contains no thrombomodulin, which means protein C is never switched on, and the brakes are never tested. It ignores platelets, von Willebrand factor, factor eight, and the whole fibrinolytic system.

So what is the INR good for? It is an excellent prognostic marker. It is part of the MELD and PELD scores used to rank patients for transplant. It is part of the entry criteria for paediatric acute liver failure. It is part of the LIU score, which we will meet later. And it is a fast marker of whether the liver's synthetic function is improving or getting worse. It is also, of course, the right test for monitoring warfarin.

What the INR is not is a measure of bleeding risk in liver disease. It is not a target to correct with plasma before a procedure. And it cannot be compared fairly between two different laboratories. Use the INR as a prognostic score, not as a test of haemostasis.

Part five. Same blood, two results.

How big is the difference between laboratories? A study by Trotter and colleagues, published in 2004, answered this very directly. They took blood from twenty-nine patients on the liver transplant waiting list, all at a single blood draw, and sent the samples to three clinical laboratories.

On the same samples, the mean INR was one point four in one laboratory and one point nine in another. Because the INR is part of the MELD score, the mean MELD score changed from about thirteen and a half to about seventeen. That moved a typical patient from the fifty-eighth to the seventy-seventh centile of waiting-list priority. Nothing about the patients had changed. Only the reagent was different.

A version of the INR calibrated on plasma from patients with cirrhosis, sometimes called the INR liver, removes most of this variation. But it is not in routine use. The PELD score and the definitions of paediatric acute liver failure have exactly the same problem. So when you record an INR in a child with liver disease, record which laboratory measured it, and compare trends within the same laboratory.

Part six. Vitamin K deficiency, liver failure, or consumption?

A long INR in a child has three common causes. The first is vitamin K deficiency. The second is failure of the liver cells. The third is consumption, meaning disseminated intravascular coagulation, or DIC. These three can be separated with two or three factor assays and one injection of vitamin K.

In vitamin K deficiency, factor seven is low, and it is the first factor to fall because it has the shortest half-life. But factor five is normal, because factor five does not need vitamin K. Factor eight is normal. Fibrinogen is normal. A test called PIVKA two is high. PIVKA stands for protein induced by vitamin K absence, and PIVKA two is the unfinished form of prothrombin that the liver releases when it has no vitamin K. Most important of all, after vitamin K is given by injection, the INR corrects within hours to a day. The typical settings are cholestasis, a breastfed infant who has not had adequate prophylaxis, cystic fibrosis, long courses of antibiotics, and any cause of fat malabsorption.

In liver cell failure, factor seven is low, and factor five is low as well, because the liver cells can no longer make it. Factor eight is normal or high, because it comes from endothelium. Fibrinogen is usually preserved until late. Giving vitamin K makes no meaningful difference to the INR. The typical settings are acute liver failure and decompensated chronic liver disease.

In DIC, the factors are being used up faster than they can be made. So factor seven, factor five and factor eight are all low, and fibrinogen is low and falling. PIVKA two is not helpful here. Vitamin K does not correct the INR. The typical settings are sepsis, shock and extensive tissue necrosis.

Now look at the three factors together. Factor seven depends on vitamin K and has the shortest half-life, so it is the most sensitive marker, but also the least specific: it falls in all three conditions. Factor five does not depend on vitamin K, so it falls only when the liver cells fail or when factors are being consumed. Factor eight is not made by the liver cell, so it rises in liver failure and falls in DIC.

This gives a simple reading. If factor seven is low and factor five is normal, think of vitamin K deficiency. If factor seven and factor five are both low and factor eight is high, think of liver failure. If factor five and factor eight are both low, think of consumption. The answer comes from the pattern, not from any single number. So order factors five, seven and eight together.

Part seven. Factor five and the French prognostic school.

Because factor five is made only by liver cells and does not need vitamin K, it is a pure readout of how well the liver cells are working. French hepatologists and surgeons built their approach to acute liver failure around it.

The best-known example is the Clichy criteria, from the Paul Brousse hospital near Paris, summarised by Bismuth and colleagues in 1995. In a patient with fulminant hepatitis and confusion or coma, a factor five level below twenty percent in a patient younger than thirty, or below thirty percent in a patient aged thirty or older, identified those who needed a transplant. Of one hundred and thirty-nine patients who met these criteria, only one recovered without a transplant.

A second study came from King's College Hospital in London, published by Pereira and colleagues in 1992, in patients with paracetamol overdose. An admission factor five of ten percent or less had a sensitivity of ninety-one percent for death. And a ratio of factor eight to factor five above thirty was seen in ten of eleven patients who died, and in none of the survivors of paracetamol poisoning. The prothrombin time on admission, in contrast, did not usefully predict the outcome. The lesson is that factor five tells you about the liver cell directly, while the INR mixes the liver cell with factor seven and with vitamin K status.

What about children? A study by Grama and colleagues, published in 2022, looked at forty children with acute liver failure, defined by the criteria of the Pediatric Acute Liver Failure Study Group. They asked how well each test predicted death or the need for transplant. The area under the curve was zero point eight nine for factor seven, zero point eight two for factor five, and zero point eight six for the peak INR. They also compared the two classic sets of criteria. The King's criteria had a sensitivity of ninety-three percent, but a specificity of only fifty-four percent. The Clichy criteria had a specificity of ninety-five percent, but a sensitivity of only fifty percent. In other words, the Clichy criteria were right when they said a child would die, but they missed half of the children who did. The King's criteria caught almost all of them, but also flagged many children who would have survived.

It helps to remember how paediatric acute liver failure is defined. The child must have an INR of one point five or more with encephalopathy, or two point zero or more without encephalopathy, and the INR must not correct with vitamin K. Encephalopathy is required at a lower INR because it is so hard to recognise in small children. The LIU score, described by Lu and colleagues in 2013, combines the peak bilirubin, INR and ammonia over time, and reached a concordance index of about zero point eight one. The key message is that no single value on a single day can replace watching the trend.

Part eight. Vitamin K in infants.

This part is where paediatric hepatology differs most from adult practice. In adults, vitamin K deficiency is a minor issue. In infants, it can be the first sign of liver disease, and it can be fatal.

Vitamin K deficiency bleeding in infants is traditionally divided into three forms. Early bleeding occurs in the first day of life, usually because the mother took drugs that interfere with vitamin K. Classic bleeding occurs in the first week. Late bleeding occurs after the first week, with a peak between three and eight weeks of age. Late bleeding is the dangerous form, because it is usually intracranial. In Europe, without any prophylaxis, it occurs in about four to seven of every one hundred thousand births. Breast milk contains very little vitamin K, so breastfed babies are most at risk.

The babies who bleed late, despite prophylaxis, are very often babies with undiagnosed cholestasis. The clearest data come from van Hasselt and colleagues, who published in 2008 a study of infants with biliary atresia from Dutch and Danish registries. They asked how many of these infants had already had a vitamin K deficiency bleed by the time biliary atresia was diagnosed.

At that time the Netherlands used a regimen of one milligram of oral vitamin K at birth, followed by a very small daily oral dose of twenty-five micrograms. Among breastfed infants with biliary atresia on this regimen, twenty-five of thirty had bled, which is about eighty-three percent. Among breastfed infants in Denmark who had received two milligrams intramuscularly at birth, about ten percent had bled. Among breastfed infants who received one milligram orally every week, about eight percent had bled. And among formula-fed infants, the figure was about one percent, because formula contains added vitamin K.

A second study from the same group looked at infants with alpha one antitrypsin deficiency. Fifteen of twenty breastfed infants had a bleed, compared with none of twenty formula-fed infants. The risk did not depend on how high the conjugated bilirubin was.

The message is that in these infants, the bleed is often the first sign that anything is wrong with the liver. A late vitamin K deficiency bleed should be seen as a failure to detect cholestasis in time.

Other groups remain at risk even after the diagnosis is made. In cholestatic infants fed a hydrolysed whey formula, the risk of vitamin K deficiency was about twenty-five times higher than on a regular formula. After a Kasai portoenterostomy for biliary atresia, between ten and twenty-two percent of children remained vitamin K insufficient despite a water-soluble multivitamin

preparation plus extra vitamin K. In the North American ChiLDReN network, deficiency of at least one fat-soluble vitamin was almost universal when the total bilirubin stayed at two milligrams per decilitre or above after the operation.

PIVKA two is useful here. Because it reflects the unfinished prothrombin made during the deficiency, it remains raised for several days after vitamin K has been given. So it can confirm the diagnosis even when a baby has already received treatment.

How do different countries give vitamin K at birth? The European society, ESPGHAN, in its 2016 statement, recommends one milligram intramuscularly at birth. As alternatives, it accepts two milligrams orally at birth, at four to six days and at four to six weeks, or two milligrams orally at birth followed by one milligram weekly for three months. It prefers the intramuscular route, and it says oral regimens are unsuitable for infants with cholestasis, preterm infants, and infants with malabsorption. The American Academy of Pediatrics, in 2003, recommended a single intramuscular dose of half a milligram to one milligram. There is no licensed oral product in the United States.

The Netherlands later raised its daily oral dose sixfold, from twenty-five to one hundred and fifty micrograms. After this change, intracranial vitamin K deficiency bleeding fell from about three to about one per one hundred thousand births. In the United Kingdom, where units chose between intramuscular and oral prophylaxis, a national surveillance study from 2006 to 2008 found eleven cases of vitamin K deficiency bleeding. Three occurred after intramuscular prophylaxis, and two of those three infants had biliary atresia.

So every oral regimen leaves a small number of cases, and that residue is largely made up of infants with cholestasis that has not yet been recognised. For a cholestatic infant, or one you suspect is cholestatic, give vitamin K by injection as soon as you suspect it. Do not rely on the prophylaxis given at birth.

Part nine. Rebalanced haemostasis.

For many years, doctors believed that patients with liver disease were naturally anticoagulated. The INR was high, so they must be at risk of bleeding and protected from clotting. We now know this is wrong.

Go back to the map in part one. The liver makes the accelerator and the brakes, and in liver disease it loses both. On the side that pushes towards bleeding, there is a low platelet count, poorly functioning platelets, low levels of the liver-made clotting factors, low or dysfunctional fibrinogen, low levels of the fibrinolysis inhibitors alpha two antiplasmin, factor thirteen and TAFI, and a high level of tissue plasminogen activator. On the side that pushes towards clotting, there is a

very high level of von Willebrand factor with low ADAMTS thirteen, a high factor eight, low antithrombin, protein C and protein S, low plasminogen, and a high level of PAI one.

The result is a new balance, set at a lower level. In a stable patient it works. But it is fragile, and it can be tipped either way by infection, kidney failure or rising portal pressure. And it explains why patients with liver disease can develop portal vein thrombosis and venous thromboembolism while their INR is well above normal.

The evidence comes from tests that include the brakes. Tripodi and colleagues, in 2005, measured thrombin generation in patients with cirrhosis, adding thrombomodulin so that protein C could be switched on. Under these conditions, patients with cirrhosis generated as much thrombin as healthy controls. In a later study by Rassi and colleagues in 2020, ninety-four percent of patients with cirrhosis had normal or high thrombin generation before any plasma was given.

The same is true in acute liver failure. Stravitz and colleagues, in 2012, studied fifty-one adults with acute liver injury or failure. Their mean INR was three point four, with a range from one and a half to nearly ten. Yet sixty-three percent of them had completely normal values on all five parameters of thromboelastography. Clot strength actually rose with the severity of liver injury, in parallel with factor eight. A weak clot was seen only in patients whose platelet count was below about one hundred and twenty-six.

The conclusion is that bleeding in liver disease is mostly caused by portal hypertension and by mechanical injury during procedures, not by a shortage of clotting factors.

What about children? The paediatric evidence agrees, but it is small and comes mostly from laboratory studies at single centres. In Stockholm, Magnusson and colleagues studied sixty-three children with chronic liver disease. Children with normal routine tests had normal thrombin generation. In children with abnormal tests, both the procoagulants and the anticoagulants were low. In Melbourne, Beattie and colleagues studied forty-two children and found that those with a PELD score above fifteen had lower prothrombin conversion and slower thrombin decay, and that children with portal vein obstruction behaved differently from those with disease of the liver tissue itself.

Two studies looked at viscoelastic tests during paediatric liver transplantation. In Seoul, Cho and colleagues analysed seven hundred and thirty-one ROTEM tests in one hundred and six children. A maximum clot firmness of thirty millimetres on the EXTEM test identified a platelet count below thirty very accurately, with

an area under the curve of zero point nine eight five. A maximum clot firmness of six millimetres on the FIBTEM test identified a fibrinogen below one hundred, with an area under the curve of zero point eight eight. But in Houston, Deshotels and colleagues studied seventy-two transplant recipients and asked which tests predicted blood loss. Standard tests did better than ROTEM, and the only single test that predicted bleeding was the platelet count. So in children, viscoelastic tests reliably reflect platelets and fibrinogen, but they do not yet predict who will bleed.

Part ten. What plasma really does.

If the INR does not measure bleeding risk, what happens when we give plasma to correct it?

Youssef and colleagues, in 2003, looked at patients with liver disease who received between two and six units of fresh frozen plasma. Only ten to twelve percent had their prothrombin time corrected to within three seconds of normal. Rassi and colleagues, in 2020, measured thrombin generation before and after a standard dose of plasma in fifty-three patients. On average it rose by less than six percent. In about a third of patients, thrombin generation was actually lower after the plasma than before. And only one of the fifty-three moved from low to normal thrombin generation.

There are good reasons for this. Plasma contains the brakes as well as the accelerators, so it replaces protein C along with the clotting factors, and the balance barely moves. Plasma also adds volume to a circulation that is already under high portal pressure, which may make bleeding from varices more likely. And it carries the risks of any blood product, including transfusion-associated circulatory overload and transfusion-related acute lung injury.

So plasma given before a procedure treats the INR, not the patient.

Part eleven. Who actually bleeds.

The largest study of this question is the PROC-BLeeD study, published by Intagliata and colleagues in 2023. It followed patients with cirrhosis at twenty centres who had procedures while in hospital: three thousand and six procedures in one thousand one hundred and eighty-seven patients.

Bleeding was uncommon. It occurred after three percent of procedures, and major bleeding after less than one percent. The strongest predictor was the type of procedure: a high-risk procedure increased the odds of bleeding about four and a half times. Severity of liver disease, measured by the MELD score, and body mass index also mattered. But the INR, the platelet count and the use of antithrombotic drugs did not predict bleeding. Interestingly, patients who bled

were more likely to have received a prophylactic transfusion beforehand, nineteen percent compared with seven percent.

A meta-analysis by Kovalic and colleagues in 2020, covering more than thirteen thousand patients, came to the same conclusion. The link between the pre-procedure INR and bleeding was weak and not statistically significant.

There is one exception worth knowing. In critically ill patients with cirrhosis, a study by Drolz and colleagues in 2016 found that very abnormal values did predict major bleeding: a platelet count below thirty, a fibrinogen below sixty milligrams per decilitre, or an activated partial thromboplastin time above one hundred seconds. So it is the extreme tail of the distribution that matters, not the usual range of abnormal results. The procedure, the severity of disease and the skill of the operator predict bleeding. The INR does not.

If not the INR, what should guide transfusion? Three randomised trials have compared transfusion guided by viscoelastic tests, meaning thromboelastography or ROTEM, with transfusion guided by the INR and platelet count. In the trial by De Pietri and colleagues in 2016, in adults with cirrhosis undergoing procedures, about seventeen percent of patients in the viscoelastic group received blood products, compared with all patients in the standard group. In the trial by Kumar and colleagues in 2020, in adults with non-variceal bleeding, the figures were about twenty-seven percent and eighty-seven percent. And in 2026, Verma and colleagues published the only paediatric trial, in seventy-six children with an INR between two and three and a half or a platelet count between twenty and fifty. About seventy-six percent of children in the viscoelastic group were transfused, compared with all children in the standard group. The volumes given were similar, and costs were higher.

In all three trials, fewer patients were transfused and there was no more bleeding. The real value of viscoelastic testing is that it gives the proceduralist permission not to transfuse.

Part twelve. What the guidelines say.

Four documents are most relevant: the American Association for the Study of Liver Diseases guidance from 2020, the European Association for the Study of the Liver guideline from 2022, the American Gastroenterological Association update from 2019, and the North American paediatric society's position paper on paediatric acute liver failure from 2022.

On the INR before procedures, all four agree. The INR is not a tool for predicting bleeding, and it should not be corrected with plasma before a procedure. They

also agree that viscoelastic tests have no validated targets and cannot yet predict procedural bleeding, and that tranexamic acid should not be used routinely.

On platelets, they differ. The American gastroenterology update suggests a platelet count above fifty for active bleeding or a high-risk procedure. The European guideline suggests deciding case by case when the count is below fifty and the procedure is high risk. The American liver guidance advises against routine correction. They also differ on thrombopoietin receptor agonists, which raise the platelet count over about ten days. They are an alternative to platelet transfusion in the gastroenterology update, a case by case option in the European guideline, and not for routine use in the American liver guidance.

On fibrinogen, the difference is material. The gastroenterology update suggests keeping fibrinogen above one hundred and twenty milligrams per decilitre during active bleeding. The American liver guidance notes that a fibrinogen below one hundred is linked to bleeding, but that a causal link is unproven. The European guideline does not recommend correcting fibrinogen to reduce procedural bleeding.

The paediatric position paper adds two points that the adult documents do not cover. Vitamin K should be given subcutaneously or intravenously, not by mouth. And the INR used to define paediatric acute liver failure is the INR measured after vitamin K has been given.

Let me put the practical approach together. First, give vitamin K by injection before calling any INR a sign of liver failure, and recheck the INR the next day. Second, if you need to know whether the liver cells are failing, measure factor five, together with factors seven and eight. Third, look at fibrinogen. It is the one routine number that tracks real bleeding. A low fibrinogen in patients with acute-on-chronic liver failure predicts bleeding after paracentesis. Replace it for active bleeding, not for a number. Fourth, remember that these patients can clot. Portal vein thrombosis can occur with an INR of two. A raised INR is not a reason to withhold thromboprophylaxis, and plasma and prothrombin complex concentrate add to the risk of thrombosis. Fifth, look for the things that tip the balance towards bleeding. Infection releases heparin-like substances into the blood, and kidney failure impairs platelet function. Treat these before a procedure, rather than treating the INR.

To summarise in one sentence: vitamin K status first, factor five second, the patient third, and the INR only for prognosis.

Part thirteen. Four clinical cases.

Case one. A seven-week-old breastfed girl is brought in with vomiting and a bulging fontanelle. A scan shows an intracranial haemorrhage. Her INR is over five. On examination she is mildly jaundiced, and her mother says the stools have been pale for a few weeks. She received oral vitamin K at birth. What is the most likely explanation, and what should you do? This is late vitamin K deficiency bleeding, and the underlying cause is almost certainly cholestasis, with biliary atresia at the top of the list. Give vitamin K intravenously at once, and consider prothrombin complex concentrate because the bleed is life threatening. Then measure the conjugated bilirubin and refer urgently to a paediatric hepatologist. The bleed is the first sign of her liver disease.

Case two. A four-year-old boy with cystic fibrosis is admitted for a chest infection. He has had several courses of antibiotics. His INR is two point one. His ALT is mildly raised, his albumin is normal, and he looks well. Is this liver failure? Probably not. Order factors five, seven and eight, and give vitamin K by injection. If factor seven is low and factor five is normal, and the INR corrects the next day, the problem was vitamin K deficiency from malabsorption and antibiotics, not failure of the liver cells.

Case three. A nine-year-old girl has acute liver failure of unknown cause. After intravenous vitamin K, her INR is still three point five. Her factor five is fifteen percent, factor seven is ten percent, and factor eight is two hundred percent. Her fibrinogen is normal. The team wants to place a central line and asks for plasma first, to bring the INR down. What do you advise? The pattern of low factors five and seven with high factor eight confirms liver cell failure, not consumption or vitamin K deficiency. The INR is a prognostic marker here, and it is also a key part of her transplant assessment. Plasma will rarely normalise it, will add volume, and will hide the trend you need to follow. Place the line under ultrasound guidance by an experienced operator without plasma. Check the platelet count and fibrinogen, and correct them only if they are extremely low or if she bleeds.

Case four. A thirteen-year-old boy with cirrhosis from autoimmune hepatitis has an INR of one point eight and a platelet count of sixty. He develops abdominal pain, and an ultrasound shows a new portal vein thrombosis. A resident is surprised, because the INR suggested he was anticoagulated. How do you explain this? Liver disease lowers both the clotting factors and the natural anticoagulants, while factor eight and von Willebrand factor rise. The balance is fragile and can tip towards thrombosis. The INR does not measure the brakes, so it cannot tell you that he is protected from clotting. A raised INR is not a

contraindication to anticoagulation, and the decision should be made by the hepatology team on its own merits.

Key points to remember.

The liver makes the clotting factors, the natural anticoagulants, and most of the fibrinolytic proteins. In liver disease it loses both sides. Factor eight and von Willebrand factor come from endothelium and rise.

Factor seven has a half-life of about five hours, so the INR moves within hours. Albumin has a half-life of about three weeks. The INR is the fast clock and albumin is the slow clock.

Newborns have about half the adult level of both the clotting factors and the natural anticoagulants. Judge a newborn's INR against newborn ranges.

The INR was built to monitor warfarin. It is a good prognostic score, but it does not measure bleeding risk in liver disease, and it differs between laboratories.

Order factors five, seven and eight together. Seven low with five normal means vitamin K deficiency. Seven and five low with eight high means liver failure. Five and eight low means consumption.

Factor five is a pure measure of the liver cell. In children with acute liver failure, the Clichy criteria are specific but miss half the deaths, and the King's criteria are sensitive but over-call.

Late vitamin K deficiency bleeding peaks at three to eight weeks and is usually intracranial. It is often the first sign of cholestasis. Oral prophylaxis fails in cholestatic infants.

Give vitamin K by injection, not by mouth, to any infant you suspect is cholestatic, and before you call any INR a sign of liver failure.

Haemostasis in liver disease is rebalanced at a lower level. Patients can bleed, but they can also clot, even with a high INR.

Plasma rarely corrects the INR and barely changes thrombin generation. Do not give it to fix a number before a procedure.

Procedural bleeding is predicted by the type of procedure and the severity of disease, not by the INR. Only extreme values of platelets, fibrinogen and the activated partial thromboplastin time matter.

Viscoelastic testing reduces transfusion without increasing bleeding, including in children.

Questions to test yourself.

Question one. Why can a child with cholestasis have a long INR even though the liver cells are healthy? Answer: vitamin K is fat soluble and needs bile to be absorbed. Without it, the liver cannot finish factors two, seven, nine and ten, so the INR rises even though the liver cells are working normally.

Question two. Why does the INR change so quickly when the liver fails, while albumin changes slowly? Answer: the INR is driven mainly by factor seven, which has a half-life of about five hours. Albumin has a half-life of about three weeks.

Question three. The same blood gives an INR of one point four in one laboratory and one point nine in another. Why? Answer: the INR is calibrated for patients on warfarin, and different reagents respond differently to the clotting pattern of liver disease. Always compare INR values from the same laboratory.

Question four. A child has a long INR. Factor seven is low, factor five is normal and factor eight is normal. What is the diagnosis, and what test confirms it? Answer: vitamin K deficiency. The INR corrects within a day of vitamin K given by injection, and PIVKA two is raised.

Question five. Why does factor eight rise in liver failure but fall in DIC? Answer: factor eight is made mainly by endothelial cells, not by liver cells, so it is preserved or even raised when the liver fails. In DIC it is consumed along with the other factors.

Question six. A breastfed infant with biliary atresia received oral vitamin K at birth. Is the infant protected from vitamin K deficiency bleeding? Answer: no. Oral prophylaxis is unreliable in cholestasis. The infant needs vitamin K by injection as soon as cholestasis is suspected.

Question seven. Why does giving plasma to a patient with cirrhosis barely improve thrombin generation? Answer: plasma contains the natural anticoagulants as well as the clotting factors, so it replaces both sides of the balance. It also adds volume and raises portal pressure.

Question eight. In the PROC-BLeeD study, what predicted bleeding after procedures in cirrhosis? Answer: the type of procedure, the severity of liver disease measured by MELD, and body mass index. The INR and platelet count did not.

End of chapter.